Joint Clutter Suppression and Moving Target Indication in 2-D Azimuth Rotated Time Domain for Single-Channel Bistatic SAR

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***Abstract*— Moving target indication (MTI) in bistatic synthetic aperture radar (BiSAR) is a promising task in both civilian and military fields. However, it suffers severe challenges under the influence of clutter. MTI in BiSAR mainly faces three challenges: 1) the range cell migration (RCM) of BiSAR is larger than that of monostatic synthetic aperture radar; 2) the clutter range-Doppler spectrum is severely extended; and 3) the clutter characteristic is closely related to geometrical configuration, with nonhomogeneity and nonstationarity. To solve these problems, based on single-channel BiSAR, a joint clutter suppression and MTI method is proposed. First, to ensure that the energy of an arbitrary target is concentrated in one range cell, RCM correction (RCMC) is completed by the first-order keystone transform (KT) and high-order RCMC. Then, an important step named increased dimension rotation (IDR) is applied, which mainly consists of two stages. One is increased dimension processing, which introduces a 2-D azimuth rotated time domain (ARTD). The other is rotation processing, which makes signals in one range cell rotated to 2-D ARTD. After that, the distribution characteristic between the moving target and stationary clutter in 2-D ARTD is analyzed. Next, an optimal filter for clutter suppression is designed according to the characteristics of signal distribution in 2-D ARTD. Furthermore, the corresponding inverse IDR processing and fractional Fourier transform (FrFT) are performed, and finally, MTI can be realized accurately. Generally, the proposed method has good robustness and low system complexity, and its effectiveness is proven by numerical simulations.**

***Index Terms*— 2-D azimuth rotated time domain (2-D ARTD), bistatic synthetic aperture radar (BiSAR), clutter suppression, increased dimension rotation (IDR), moving target indication (MTI), single-channel.**

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I. Introduction

SYNTHETIC aperture radar (SAR) has been a research highlight whether in military or civilian fields in recent years since its ability to work in the entire day and all weather [1], [2]. In general, based on system configuration, SAR systems have two modes: monostatic SAR and bistatic SAR (BiSAR) [3]. Compared with monostatic SAR, BiSAR has a wide reconnaissance range and powerful anti-interference advantages, benefiting from the separation of transmitter and receiver [4]. With increased demands for surveillance systems, higher requirements are put forward for SAR moving target indication (MTI). Indication, imaging, localization, and estimation of velocity parameters for moving targets have attracted lots of attention and studies for the SAR community [5], [6].

Effective clutter suppression becomes critical for MTI. Existing clutter suppression methods can be mainly classified into two modes [7]: single-channel methods and multichannel methods. Multichannel methods have strict channel consistency requirements [8], [9], high system complexity, and large computational burden [7], resulting in limited applications. Therefore, improving the MTI of single-channel SAR systems is more reliable since it has lower system complexity and computational burden. Representative single-channel methods include Doppler filtering [10], Doppler velocity filtering [11], and detection methods based on parameter estimation [12]. Unfortunately, the above single-channel methods are all based on a monostatic SAR system and cannot be directly applied to BiSAR.

In BiSAR, because of its system characteristic and special geometry configuration, the main challenges of MTI can be summarized in the following three categories.

1) The larger range cellmigration (RCM) compared to monostatic SAR. It causes moving target energy dispersed in many range cells, resulting in a decrease in the signal-to-clutter and noise ratio (SCNR).
2) The extension of clutter range-Doppler (RD) spectrum caused by the motion of bistatic platforms [13]. Spectrum extension will lead to the failure of some Doppler filtering [10] methods with single-channel.

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3) The clutter is nonhomogeneous and nonstationary [14], [15], [16], [17], which is closely associated with geographic features and system geometrical configuration. It may cause degraded performance or even failure of some maximum likelihood-based [12] and space–time distribution characteristic-based detection methods, i.e., space–time adaptive processing (STAP) [18], [19], [20], [21], [22], [23].

Generally, most single-channel MTI methods are based on monostatic SAR. Freeman and Currie [24] proposed a clutter filtering method in the RD domain. Nevertheless, pulse repetition frequency (PRF) had to be at least four times higher than the clutter bandwidth to realize highly accurate detection. Another MTI method [25] based on the Doppler modulation rate was proposed. However, this method could only detect a moving target with a large azimuth velocity. Since the clutter background was invariant and the offset of the moving target was different, the multilook methods were proposed in [26] and [27]. However, high SCNR needed to be satisfied. Besides, Lv et al. [28] proposed a symmetric Doppler views method to indicate moving target under the assumption that a large azimuth time-bandwidth product was satisfied. Yang et al. [7] proposed a two-step MTI algorithm to realize moving target detection. The two-step MTI method used the subaperture image cancellation essentially, which was required to work on side-look mode. Dias and Marques [29] presented a matched filtering (MF) processing depending on a certain motion feature of a moving target. Wei et al. [30] and Zhong et al. [31] proposed an algorithm to detect moving targets by using a sharpness ratio and target shadow, which were sensitive to clutter and noise. It is obvious that, when considering moving targets with relatively strong energy, the single-channel method becomes a useful solution. Thus, the previous approaches can ensure reliable detection performance.

Comparatively, studies of BiSAR are focused on imaging theory, imaging algorithm [32], [32], [34], and so on. Only a few studies have been reported on MTI methods for BiSAR. Considering the prospective applications of MTI whether in military or civilian fields, MTI is well worthy for conducting relevant research. In [35], a mismatched compression algorithm with single-channel was applied for moving target parameters' estimation. In [36], an optimal 2-D spectrum matching algorithm was used to moving targets' imaging. Liu et al. [37] presented a keystone-based delay-correlation transform and fast searching Fourier transform (KDCT-FSFT)-based refocusing algorithm to achieve MTI. However, the aforementioned single-channel MTI methods only accumulate moving target signals but ignore the influence of strong clutter. Therefore, such indication methods will fail in strong clutter scenarios.

To cope with problems of large RCM, the extension of clutter spectrum, and geometrical configuration-dependent clutter characteristic in BiSAR, a joint clutter suppression and MTI method in 2-D azimuth rotated time domain (ARTD) for single-channel BiSAR is presented in this article. Its main innovation is to separate moving targets from stationary clutter in a newly constructed 2-D ARTD. First, the first-order keystone transform (KT) [38] is performed to remove linear range walk (RWK) regardless of unknown motion parameters. Then, high-order RCM correction (RCMC) is realized by reference compensation function [39], [40]. It is extremely difficult to achieve MTI because the moving target is submerged in clutter whether in the azimuth time domain or the Doppler domain. Thus, to achieve the separation of moving target from stationary clutter effectively, increased dimension rotation (IDR) processing is introduced, in which 2-D ARTD is constructed accompanied by IDR processing. Thereafter, the distribution characteristic of moving target and stationary clutter after rotation processing is analyzed exhaustively. Naturally, the optimal filter is constructed for clutter suppression in light of the distribution characteristic. Finally, the energy of the moving target can be accumulated after inverse IDR and fractional Fourier transform (FrFT) [42], [43]. Moreover, the corresponding position and velocity parameters can be easily obtained during IDR processing. It has been demonstrated that this method achieves better clutter suppression and MTI performance than the two-step method [7] and the Doppler filtering [10] method.

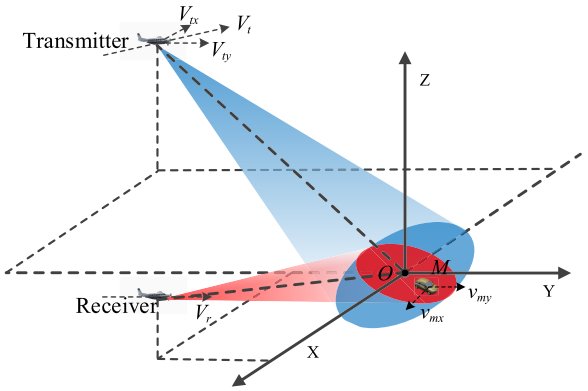


Fig. 1. Generic BiSAR geometric configuration.

This article consists of the following contents. In Section II, the moving target signal model for BiSAR is established, and RCM and RD spectrum extension are analyzed exactly. In Section III, based on single-channel BiSAR, a joint clutter suppression and MTI method is described detailedly in 2-D ARTD. Meanwhile, precise position and velocity parameters are obtained. Section IV demonstrates the effectiveness of the proposed method through some simulation cases. Section V makes a conclusion.

II. Geometry Configuration and Signal Model

The generic BiSAR geometric configuration is presented, and the moving target signal model with constant speed is also constructed in this part. Besides, RCM and RD spectrum extension characteristics are analyzed exactly.

A. Signal Model

Fig. 1 presents the generic BiSAR geometric configuration in the Cartesian coordinate system. Without losing generality, the beam center $O(0, 0)$ at azimuth zero moment is chosen as the origin, and the receiver's flight direction is parallel to the Y-axis. XOY plane represents the ground's surface, and

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the Z-axis is perpendicular to XOY plane. (X_T, Y_T, Z_T) and (X_R, Y_R, Z_R) denote the positions of the transmitter and the receiver, respectively. Velocities of two platforms are $(0, V_r, 0)$ and $(V_{tx}, V_{ty}, 0)$, where $V_t^2 = V_{tx}^2 + V_{ty}^2$. There is a moving target M in illumination scene, with (x_m, y_m) as initial position and $(\upsilon_{mx}, \upsilon_{my})$ as velocity in XOY plane. This article assumes that the BiSAR system operates in the strip mode.

The bistatic range histories $R_{\mathrm{BM}}(\eta; x_m, y_m, \upsilon_{mx}, \upsilon_{my})$ at azimuth slow time η can be easily obtained according to positions and velocities of the bistatic platform and moving target. In convenience, $(x_m, y_m, \upsilon_{mx}, \upsilon_{my})$ is denoted as $\boldsymbol{\xi}_m$. Then, bistatic range histories are represented as

$$R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m) = R_{\mathrm{RM}}(\eta; \boldsymbol{\xi}_m) + R_{\mathrm{TM}}(\eta; \boldsymbol{\xi}_m) \tag{1}$$

$$R_{\mathrm{RM}}(\eta; \boldsymbol{\xi}_m) = \sqrt{(X_R - x_m - \upsilon_{mx}\eta)^2 + (Y_R - y_m + V_{r,my}\eta)^2 + Z_R^2} \tag{2}$$

$$R_{\mathrm{TM}}(\eta; \boldsymbol{\xi}_m) = \sqrt{(X_T - x_m + V_{t,mx}\eta)^2 + (Y_T - y_m + V_{t,my}\eta)^2 + Z_T^2} \tag{3}$$

where $V_{r,my} = V_r - \upsilon_{my}$, $V_{t,mx} = V_{tx} - \upsilon_{mx}$, and $V_{t,my} = V_{ty} - \upsilon_{my}$.

Expanding $R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m)$ into the Taylor series at azimuth slow time $\eta = 0$ and keeping up to biquadratic terms

$$R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m) \approx R_{BM0} + k_1(\boldsymbol{\xi}_m)\eta + k_2(\boldsymbol{\xi}_m)\eta^2 + k_3(\boldsymbol{\xi}_m)\eta^3 + k_4(\boldsymbol{\xi}_m)\eta^4 \tag{4}$$

where

$$R_{BM0} = \sqrt{(X_R - x_m)^2 + (Y_R - y_m)^2 + Z_R^2} + \sqrt{(X_T - x_m)^2 + (Y_T - y_m)^2 + Z_T^2} \tag{5}$$

and

$$k_i(\boldsymbol{\xi}_m) = \frac{1}{i!}\frac{dR_{\mathrm{BM}}^i(\eta; \boldsymbol{\xi}_m)}{d\eta^i}\bigg|_{\eta=0} \tag{6}$$

where $\{k_i(\boldsymbol{\xi}_m), i = 1, 2, 3, 4\}$ is the coefficient of corresponding terms and ! represents the factorial.

Suppose that the linear frequency-modulated (LFM) pulse is transmitted, and then the received signal of the moving target after demodulation is derived as

$$S_1(\tau, \eta; \boldsymbol{\xi}_m) = \sigma_m \omega_r\left(\tau - \frac{R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m)}{c}\right)\omega_a\left(\frac{\eta - \eta_r}{T_s}\right) \times \exp\left[-j2\pi f_c \frac{R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m)}{c}\right] \times \exp\left[j\pi K_r\left(\tau - \frac{R_{\mathrm{BM}}(\eta; \boldsymbol{\xi}_m)}{c}\right)^2\right] \tag{7}$$

where σ_m is the complex scattering coefficient, η_r is the beam center moment, T_s is the synthetic aperture time, K_r denotes the frequency modulation rate of the transmitted LFM signal, and τ indicates the fast time. c and f_c represent the speed of light and carrier frequency, respectively. ω_a and ω_r denote the azimuth and range envelopes. $\lambda = c/f_c$ represents the wavelength.

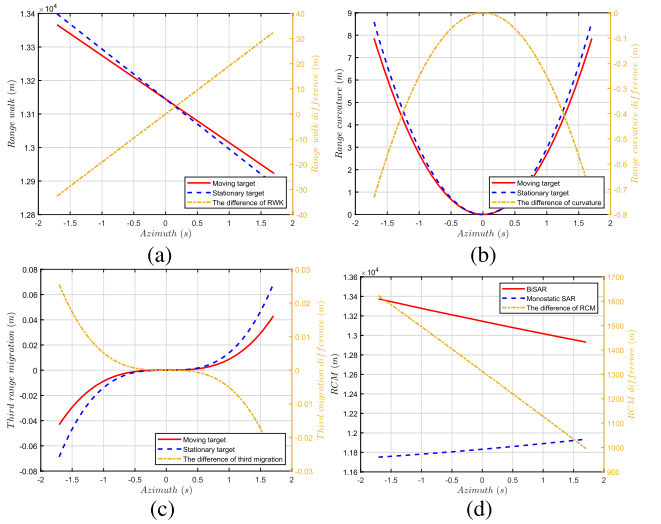


Fig. 2. RCM characteristic difference between moving and stationary target, and RCM comparison between BiSAR and monostatic SAR. (a) RWK. (b) RC. (c) Third-order range migration. (d) Moving target RCM comparison of BiSAR and monostatic SAR.

B. RCM Characteristic Analysis

This section analyzes RCM characteristics of moving and stationary targets for BiSAR, and compares RCM differences between monostatic and BiSAR systems.

The corresponding simulation parameters are in accordance with Table I, and the velocity of the moving target is set to be (10, 10) m/s. The red solid line represents the RCM of the moving target, the blue dashed line indicates the RCM of the stationary target, and the orange dotted-dashed line denotes the RCM difference in Fig. 2(a)–(c). Fig. 2(a)–(c) presents range walk RWK, range curve (RC), and the third-order range migration, in which the expansion is about 550, 9, and 0.14 m, respectively. In addition, the differences in RWK, RC, and the third-order range migration between the moving target and the stationary target are about 60, 0.7, and 0.04 m. Therefore, the RWK is dominant, the RC cannot be ignored because it is higher than one range resolution unit, and the third-order range migration can be ignored because it is less than one range resolution unit. In Fig. 2(d), it indicates the moving target RCM difference between BiSAR and monostatic SAR. From the difference curves (marked as an orange dotted-dashed line), a large RCM difference is demonstrated. The above analysis of RCM characteristics is the foundation for the subsequent RCMC.

C. Doppler Spectrum Extension Characteristic Analysis

This section derives the RD spectrum of the moving target and the stationary target, and describes the extension characteristic of clutter RD spectrum in BiSAR in detail. The Doppler frequency of stationary clutter equals zero, and the Doppler frequency of the moving target will not be zero when bistatic platforms are stationary, so separating the moving target from

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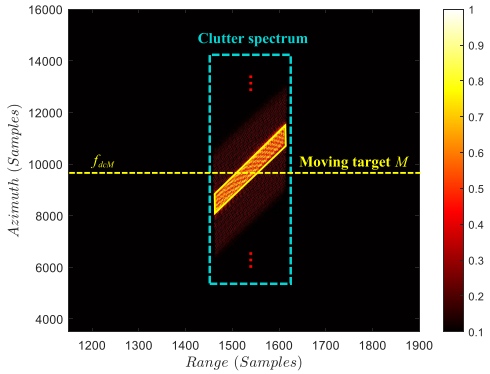


Fig. 3. Moving target RD spectrum is submerged by that of clutter.

the clutter in the Doppler domain is a breeze. However, when the platform is in motion, the Doppler frequency of stationary clutter is no longer zero and changes with the relative angle. It causes the extension of stationary clutter Doppler spectrum. When the velocity of the moving target is fairly small, it is prone to be drowned in stationary clutter Doppler spectrum, which leads to great difficulty in MTI.

To further analyze the corresponding RD spectrum, the 2-D spectrum of the moving target is acquired via a method of series reversion (MSR) [45], [46] first. After performing the Taylor series expansion at $f_\tau = 0$ and keeping up to quadratic items, it yields

$$\begin{aligned}\phi_1(f_\eta, f_\tau; \boldsymbol{\xi}_m) = {} & \phi_{\text{res}} + \phi_{rg}(f_\tau; \boldsymbol{\xi}_m) + \phi_{az}(f_\eta; \boldsymbol{\xi}_m) \\ & + \phi_{\text{rcm}}(f_\eta; \boldsymbol{\xi}_m) f_\tau + \phi_{\text{src}}(f_\eta; \boldsymbol{\xi}_m) f_\tau^2 \end{aligned} \quad (8)$$

where $\phi_{rg}(f_\tau; \boldsymbol{\xi}_m)$ is the range modulation term, $\phi_{az}(f_\eta; \boldsymbol{\xi}_m)$ denotes the azimuth modulation term, $\phi_{\text{rcm}}(f_\eta; \boldsymbol{\xi}_m)$ indicates the RCM term, and $\phi_{\text{src}}(f_\eta; \boldsymbol{\xi}_m)$ represents the secondary range compression (SRC) term. ϕ_{res} is the residual phase that does not depend on the azimuth frequency f_η and the range frequency f_τ. The detailed analytical derivation and expressions of the above terms can be found in [46].

The term $\phi_{rg}(f_\tau; \boldsymbol{\xi}_m)$ can be compensated for after range compression, and then, the principle of stationary phase (POSP) is used to derive RD spectrum based on 2-D spectrum (8), which is given by

$$\begin{aligned}\phi_2(f_\eta, \tau; \boldsymbol{\xi}_m) = {} & \phi_{az}(f_\eta; \boldsymbol{\xi}_m) + \phi_{\text{res}} - \frac{\phi_{\text{rcm}}(f_\eta; \boldsymbol{\xi}_m)}{4\phi_{\text{src}}(f_\eta; \boldsymbol{\xi}_m)} \\ & - \frac{\pi \phi_{\text{rcm}}(f_\eta; \boldsymbol{\xi}_m)}{4\phi_{\text{src}}(f_\eta; \boldsymbol{\xi}_m)} \tau - \frac{\pi^2}{\phi_{\text{src}}(f_\eta; \boldsymbol{\xi}_m)} \tau^2. \end{aligned} \quad (9)$$

Suppose that there is a stationary target P, and homologous parametric variable is represented as $\boldsymbol{\xi}_p := (x_p, y_p, 0, 0)$ to be in accordance with the expression $\boldsymbol{\xi}_m$. Then, the bistatic range histories' coefficients of P after Taylor series expansion are $\{k_i(\boldsymbol{\xi}_p), i = 1, 2, 3, 4\}$.

By variable substitution $\boldsymbol{\xi}_m \to \boldsymbol{\xi}_p$ while keeping the derivation process unchanged, 2-D spectrum and RD spectrum of P can be spontaneously obtained as $\phi_1(f_\eta, f_\tau; \boldsymbol{\xi}_p)$ and $\phi_2(f_\eta, \tau; \boldsymbol{\xi}_p)$. Fig. 3 demonstrates that the moving target's RD spectrum is submerged by that of clutter.

The RD spectrum of several stationary targets and one moving target M is drawn in Fig. 3. f_{dcM} represents Doppler centroid of moving target. It is clear that its RD spectrum has a certain extension for each target, which is primarily caused by Doppler frequency rate (DFR) and synthetic aperture time. Moreover, the clutter Doppler spectrum is further broadened by different Doppler centroids of different stationary targets. The more severely the extension of the clutter RD spectrum, the more easily the moving target is masked. In Fig. 3, moving target M is completely submerged because of the extension of the clutter RD spectrum. Then, traditional algorithms based on Doppler filtering will fail.

Through the above analysis, factors of the azimuth extension about clutter RD spectrum can be mainly summarized as follows: different Doppler centroids of different stationary targets, and the extension of each stationary target caused by the combination of DFR and synthetic aperture time. For a more explicit perception of this extension, Doppler parameters of 2 km $\times$ 2 km scene target are analyzed as follows.

The first-, second-, and third-order Doppler parameters can be expressed as

$$\begin{cases} f_{\text{dc}}(\boldsymbol{\xi}_p) & = -\frac{k_1(\boldsymbol{\xi}_p)}{\lambda} \\ f_{\text{dr}}(\boldsymbol{\xi}_p) & = -2\frac{k_2(\boldsymbol{\xi}_p)}{\lambda} \\ f_{d3}(\boldsymbol{\xi}_p) & = -6\frac{k_3(\boldsymbol{\xi}_p)}{\lambda} \end{cases} \quad (10)$$

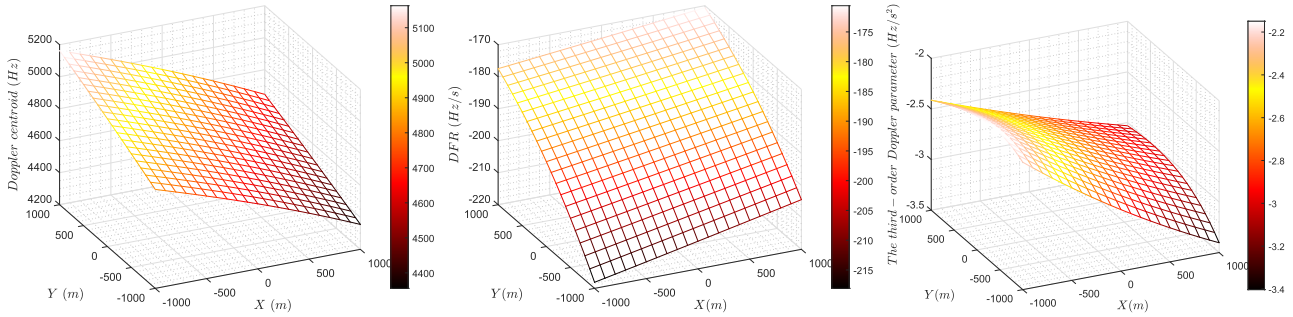
where f_{dc}, f_{dr}, and f_{d3} represent the Doppler centroid, DFR, and the third-order Doppler parameter, respectively.

In Fig. 4, Doppler parameters are illustrated in detail. The system parameters are also consistent with Table I. The Doppler centroid, DFR, and the third-order Doppler parameter span about 800 Hz, 50 Hz/s, and 1 Hz/s^2, respectively. Considering PRF = 1200 Hz, the clutter RD spectrum is almost spread over the entire Doppler frequency domain. The moving target will be easily submerged. Moreover, the third-order Doppler parameter is much less than the Doppler centroid and DFR. It also corroborates the key factors for the extension of the clutter RD spectrum.

III. Joint Clutter Suppression and Moving Target Indication in 2-D ARTD

Based on the analysis aforementioned, it is almost impossible to distinguish moving targets from stationary targets in the Doppler domain because of severe clutter spectrum extension. In this section, based on single-channel BiSAR, a joint clutter suppression and MTI method in 2-D ARTD is described exhaustively. First, the first-order KT is preformed to remove RWK, and then, the high-order RCM is corrected by the reference compensation function of scene center [35], [39]. Next, IDR processing is applied, aiming to increase the degree of freedom and construct 2-D ARTD. By analyzing the distribution characteristic in 2-D ARTD, the optimal filter is designed precisely for clutter suppression. Therefore, the effective MTI can be easily realized after inverse IDR and accumulation operation finally. Meanwhile, the beam center moment of the moving target can be also determined easily

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(a) (b) (c)

Fig. 4. Doppler parameters of 2 × 2 km scene target. (a) Doppler centroid. (b) DFR. (c) Third-order Doppler parameter.

during the IDR procedure, so as to further realize accurate positioning and velocity estimation of the moving target.

A. RCMC via Keystone Transform

The signal in the f_τ–η domain is derived after range compression

$$S_2(f_\tau,\eta;\boldsymbol{\xi}_m)=\omega_r\left(\frac{f_\tau}{B_r}\right)\omega_a\left(\frac{\eta-\eta_r}{T_s}\right)\times\exp\left[-j2\pi\frac{(f_c+f_\tau)}{c}R_{\mathrm{BM}}(\eta;\boldsymbol{\xi}_m)\right]. \quad (11)$$

Substituting (4) into (11), the phase of the range frequency domain is written as follows:

$$\phi_3(f_\tau,\eta;\boldsymbol{\xi}_m)=-\frac{2\pi}{c}(f_\tau+f_c)\left[R_{BM0}+\sum_{i=1}^{4}k_i(\boldsymbol{\xi}_m)\eta^i\right]. \quad (12)$$

Noticed that the linear coupling term between range frequency f_τ and azimuth time η, it represents RWK inherently. In order to correct RWK, the first-order KT is performed

$$\eta_k=\frac{f_\tau+f_c}{f_c}\eta \quad (13)$$

where η_k is new azimuth time variable after KT. Then, the phase term $\phi_3(f_\tau,\eta;\boldsymbol{\xi}_m)$ is changed into

$$\phi_4(f_\tau,\eta_k;\boldsymbol{\xi}_m)=-\frac{2\pi}{c}(f_\tau+f_c)R_{BM0}-\frac{2\pi}{c}\left[\begin{array}{l}k_1(\boldsymbol{\xi}_m)f_c\eta_k+\dfrac{k_2(\boldsymbol{\xi}_m)f_c^2}{f_\tau+f_c}\eta_k^2+\\ \dfrac{k_3(\boldsymbol{\xi}_m)f_c^3}{(f_\tau+f_c)^2}\eta_k^3+\dfrac{k_4(\boldsymbol{\xi}_m)f_c^4}{(f_\tau+f_c)^3}\eta_k^4\end{array}\right]. \quad (14)$$

It is obvious that the first-order coupling term between f_τ and η has been eliminated fully. Expanding the above phase term $\phi_4(f_\tau,\eta_k;\boldsymbol{\xi}_m)$ with the Taylor series at $f_\tau=0$ yields

$$\phi_5(f_\tau,\eta_k;\boldsymbol{\xi}_m)=-\frac{2\pi}{\lambda}R_{\mathrm{BM}}(\eta_k;\boldsymbol{\xi}_m)-\frac{2\pi}{c}\phi_{1\mathrm{st}}(\boldsymbol{\xi}_m)f_\tau \quad (15)$$

where

$$\phi_{1\mathrm{st}}(\boldsymbol{\xi}_m)=R_{BM0}-k_2(\boldsymbol{\xi}_m)\eta_k^2-2k_3(\boldsymbol{\xi}_m)\eta_k^3-3k_4(\boldsymbol{\xi}_m)\eta_k^4. \quad (16)$$

According to [39], the spatial variation of residual RCM is fairly small. Therefore, the compensation function for high-order RCM can be established by reference target O and yields

$$H_{\mathrm{RCM}}=\exp\left\{j\frac{2\pi}{c}\left[\phi_{1\mathrm{st}}(\boldsymbol{\xi}_0)-R_{BM0}\right]f_\tau\right\} \quad (17)$$

where $\boldsymbol{\xi}_0:=(0,0,0,0)$ is the corresponding parameter representation for reference target O.

After high-order RCMC and inverse fast Fourier transform (IFFT) of range dimension, the echo in the τ–η_k domain can be obtained, which is

$$S_3(\tau,\eta_k;\boldsymbol{\xi}_m)=A\mathrm{sinc}\left[B_r\left(\tau-\frac{R_{BM0}}{c}\right)\right]\omega_a\left(\frac{\eta_k-\eta_r}{T_s}\right)\times\exp\left[-j\frac{2\pi}{\lambda}R_{\mathrm{BM}}(\eta_k;\boldsymbol{\xi}_m)\right] \quad (18)$$

where A represents the amplitude of $S_3(\tau,\eta_k;\boldsymbol{\xi}_m)$ and B_r indicates the bandwidth of LFM. RCM has been completely compensated after the first-order KT and high-order RCMC. Fig. 12(a) and (b) in the simulation part shows the results before and after RCMC. It confirms that the energy of the moving target gathers into one range cell after RCMC, which facilitates our subsequent processing.

The RCMC processing above is mainly dedicated to large RWK in BiSAR. However, the second-order RCM is dominant in monostatic SAR, which works on the side-look or small squint mode. The RCMC processing above is no longer applicable. It is worth mentioning that the RCMC processing above can be suitable in highly squint monostatic SAR [39].

B. Increased Dimension Rotation Processing

Different targets of the same range cell have variant Doppler parameters, which further extends the clutter spectrum and makes MTI more challenging. If the effective separation of moving target and stationary clutter in a certain domain can be achieved, the suppression of stationary clutter and MTI can be realized. Aiming at the separation challenges, IDR processing is performed, which mainly includes two components: one is increased dimension and the other is rotation. The detailed steps are illustrated in the following.

1) Increased Dimension Processing: To perform increased dimension processing, a new domain κ, which is perpendicular to the plane with respect to $\eta_k-\tau$, is introduced, as illustrated in Fig. 5. The newly introduced κ domain indicates ARTD,

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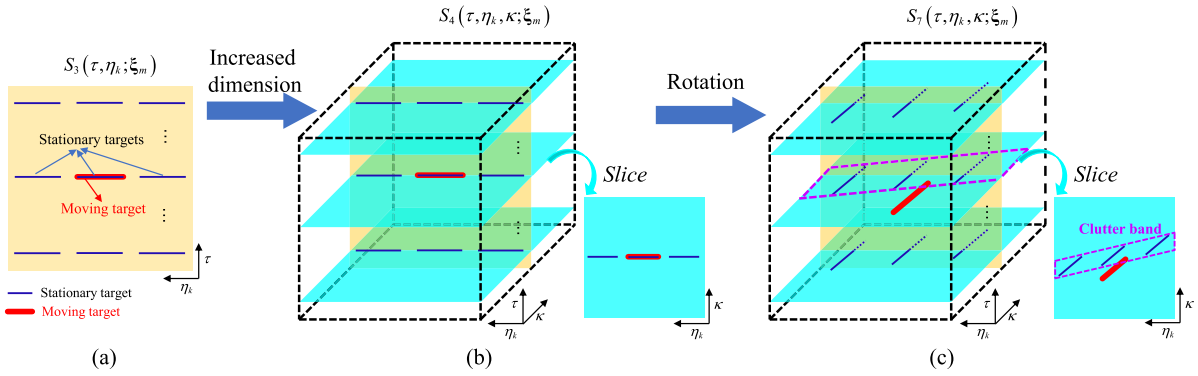


Fig. 5. IDR procedure diagrammatic sketch. (a) Echo data after RCMC. (b) Increased dimension with expanding zero matrices by both sides. (c) Rotation processing.

which will be explained in the following. It is dedicated to increasing new degree of freedom and constructing 2-D ARTD $\eta_k - \kappa$ combined with the original azimuth time domain. Next, the detailed procedure of introducing the new domain κ is elaborated on.

Suppose that the sample number of new dimension κ is Q. The original echo data after RCMC are $S_3(\tau, \eta_k; \boldsymbol{\xi}_m)$ with the size of $N_r \times N_a$, where N_r and N_a are denoted as range and azimuth dimension sampling number, respectively. The schematic of original echo data $S_3(\tau, \eta_k; \boldsymbol{\xi}_m)$ is shown in Fig. 5(a), where the yellow plane represents 2-D echo. Three range cells are selected for illustration. There are three stationary targets marked as the blue solid line in each range cell, and one of the range cells has a moving target marked as the red solid line. To perform the increased dimension, two zero matrices with size $(Q-1)/2 \times N_r \times N_a$ are established, and the original echo data $S_3(\tau, \eta_k; \boldsymbol{\xi}_m)$ are expanded into 3-D data $S_4(\tau, \eta_k, \kappa; \boldsymbol{\xi}_m)$ using the zero matrices by both sides, as shown in Fig. 5(b). The black dashed cube represents $S_4(\tau, \eta_k, \kappa; \boldsymbol{\xi}_m)$, where the yellow 2-D plane in the middle of the cube indicates original echo data $S_3(\tau, \eta_k; \boldsymbol{\xi}_m)$. The samples per dimension η_k, τ, and κ are N_a, N_r, and Q, respectively.

2) Rotation Processing: To transform the signal of the same range cell into the 2-D $\eta_k - \kappa$ domain, a specific rotation processing is applied. Fig. 5(c) depicts the graphical representation of the rotation processing, where the purple dashed line represents the clutter band. The steps of rotation processing can be summarized as follows. First, the 2-D fast Fourier transform (FFT) should be applied in the $\eta_k - \kappa$ domain, which derives

$$S_4(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m) = B\omega_a\left(\frac{f_{\eta_k} - f_{\mathrm{dc}}}{B_a}\right)\omega_\kappa\left(\frac{f_\kappa}{F_\kappa}\right)\exp\left[j\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m)\right] \quad (19)$$

where

$$B = A\,\mathrm{sinc}\left[B_r\left(\tau - \frac{R_{BM0}}{c}\right)\right]$$

$$\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m) = \left[\frac{k_1(\boldsymbol{\xi}_m)}{k_2(\boldsymbol{\xi}_m)} + 3\frac{k_1^2(\boldsymbol{\xi}_m)k_3(\boldsymbol{\xi}_m)}{4k_2^3(\boldsymbol{\xi}_m)}\right]\pi f_{\eta_k} - \left[\frac{2k_2^2(\boldsymbol{\xi}_m) + 3k_1(\boldsymbol{\xi}_m)k_3(\boldsymbol{\xi}_m)}{4k_2^3(\boldsymbol{\xi}_m)}\right]\pi\lambda f_{\eta_k}^2 + \frac{k_3(\boldsymbol{\xi}_m)\pi\lambda^2}{4k_2^3(\boldsymbol{\xi}_m)}f_{\eta_k}^3 + \phi_{\mathrm{con}}.$$

B denotes the amplitude; f_{dc} and B_a are the Doppler centroid and the Doppler bandwidth, respectively. ω_κ and F_κ represent the envelope and the sampling rate of κ dimension. $\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m)$ indicates the azimuth phase, ϕ_{con} means the constant phase of $\phi_a(f_{\eta_k}; \xi_m)$, and f_{η_k} and f_κ are frequency representations of η_k and κ.

It is necessary to explain the origin of (19). Because there is no coupling between the new established dimension κ and the azimuth dimension η_k, it is not difficult to obtain the signal $S_4(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m)$ after 2-D FFT. In $S_4(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m)$, the parameter B is independent of κ and η_k, so B can be considered as the signal amplitude. $\omega_a[(f_{\eta_k} - f_{\mathrm{dc}})/B_a]$ is the envelope in the azimuth frequency domain, and $\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m)$ is the corresponding azimuth phase, which is obtained by the Fourier transform of $\exp[-j2\pi R_{\mathrm{BM}}(\eta_\kappa; \boldsymbol{\xi}_m)/\lambda]$. It is worth noting that signal $S_4(\tau, \eta_k, \kappa; \boldsymbol{\xi}_m)$ has the value only in the middle of κ dimension. It can be considered as a unit impulse signal or an approximate sinc(•) signal along the κ dimension. Combined the condition of decoupling of the κ-axis and the η_k-axis, $\omega_\kappa(f_\kappa/F_\kappa)$ can be acquired in the $f_{\eta_k} - f_\kappa$ domain after 2-D FFT.

Second, to perform the rotation operation, the rotation phase factor is constructed as

$$\phi_{rot}(f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_r) = -\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_r)f_\kappa - \phi_{\mathrm{src}}(f_{\eta_k}; \boldsymbol{\xi}_r)f_\kappa^2 \quad (20)$$

where $\boldsymbol{\xi}_r := (x_{\mathrm{ref}}, y_{\mathrm{ref}}, 0, 0)$, and $(x_{\mathrm{ref}}, y_{\mathrm{ref}})$ is the coordinate of reference target R of the specific range cell. The middle position of the specific range cell is often chosen as the reference target. That is to say, $y_{\mathrm{ref}} = 0$ and x_{ref} can be determined by following equation [41]:

$$x = \frac{X_T}{2} + \frac{X_T C_1}{2C_2} + \frac{R_{BM0}\sqrt{C_2^2 + C_1^2 - 2C_2C_3}}{2C_2} \quad (21)$$

where

$$\begin{cases} C_1 = Y_R^2 - Y_T^2 + 2y(Y_T - Y_R) + Z_R^2 - Z_T^2 \\ C_2 = R_{BM0}^2 - X_T^2 \\ C_3 = (y - Y_R)^2 + (y - Y_T)^2 + Z_R^2 + Z_T^2. \end{cases}$$

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Next, multiplying $S_4(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m)$ by rotation phase factor, $S_5(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m)$ shown in (22), at the bottom of the page, can be obtained after simplification. According to [47], the detailed simplification process is shown as, (24), at the bottom of the next page, where $k_3(\boldsymbol{\xi}_r)$ and $k_4(\boldsymbol{\xi}_r)$ are far less than $k_1(\boldsymbol{\xi}_r)$ and $k_2(\boldsymbol{\xi}_r)$, and then, the first term of $\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_r)$ is dominant.

Subsequently, as shown in (23), at the bottom of the page, IFFT is applied in κ dimension to obtain $S_6(\tau, f_{\eta_k}, \kappa; \boldsymbol{\xi}_m)$.

Finally, the corresponding signal $S_7(\tau, \eta_k, \kappa; \boldsymbol{\xi}_m)$ after the rotation will be presented with IFFT in azimuth dimension, as shown in the following:

$$S_7(\tau, \eta_k, \kappa; \boldsymbol{\xi}_m) = \underset{f_{\eta_k}}{\mathrm{IFFT}}\big[S_6(\tau, f_{\eta_k}, \kappa; \boldsymbol{\xi}_m)\big]. \tag{25}$$

With the above rotation processing, signals in one range cell are rotated to the $\eta_k - \kappa$ domain, as shown in Fig. 5(c). Hence, the κ domain is termed ARTD.

3) Distribution Characteristic Analysis on 2-D ARTD: By the aforementioned IDR processing, the signal of one specific range cell has been rotated to the $\eta_k - \kappa$ 2-D domain. To separate moving targets from stationary clutter, it is necessary to analyze the distribution characteristic of the signal in the $\eta_k - \kappa$ domain accurately.

Suppose that there are a number of stationary targets $S_i, (i = 1, 2, \ldots, l, \ldots, L)$, and a moving target M is in one specific range cell, where L denotes the total number of stationary targets. Similarly, the expression in the $f_{\eta_k} - \kappa$ domain of stationary targets S_i can be derived according to IDR procedure, denoted as $S_6(\tau, f_{\eta_k}, \kappa; \boldsymbol{\xi}_{s_i})$. The only difference among them is the offset along the κ-axis, which is caused by the variation in the Doppler centroid of S_i, as

$$\begin{aligned}\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-s_i}) &= \frac{\pi}{2k_2(\boldsymbol{\xi}_r)}\left[\frac{k_1^2(\boldsymbol{\xi}_r)}{c} - \frac{c}{f_c^2} f_{\eta_k}^2\right] \\ &= \frac{\pi}{f_c f_{\mathrm{dr}}(\boldsymbol{\xi}_r)}\big[f_{\mathrm{dc}}^2(\boldsymbol{\xi}_r) - f_{\eta_k}^2\big].\end{aligned} \tag{26}$$

To further analyze the distribution characteristic of stationary targets and moving targets in the same range cell, three situations can be discussed as follows.

1) The coordinate of S_i coincides with reference target R. Equation $f_{\eta_k} = f_{\mathrm{dc}}(\boldsymbol{\xi}_r)$ holds at beam center moment of the corresponding target, and then, $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-s_i}) = 0$. It indicates that there is no offset along the κ dimension.
2) The coordinate of S_i do not coincide with reference target R. Equation $f_{\eta_k} = f_{\mathrm{dc}}(\boldsymbol{\xi}_{s_i}) \neq f_{\mathrm{dc}}(\boldsymbol{\xi}_r)$ holds at beam center moment of the corresponding target. There will be a certain offset along the κ dimension. Meanwhile, stationary targets in one range cell have different Doppler centroids, which implies variant offsets along the κ dimension. Next, the offset characteristic is demonstrated in Fig. 6. The corresponding simulation parameters are in accordance with Table I. This figure creates axes with the y-axis on both the left- and right-hand sides. The blue solid line and the red dashed line that belong to the left axis represent the accurate phase difference $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-s_i})$ and the first-order polynomial fitting result with respect to azimuth time η_k, respectively. The fitting error marked with the yellow dotted-dashed line, which belongs to the right axis, is much smaller than phase difference $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-s_i})$. It indicates that the first-order polynomial fitting is sufficiently accurate relative to the characterized phase difference of $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-s_i})$. Therefore, stationary targets in the same range cell will exhibit a linearly inclined distribution on the $\eta_k - \kappa$ 2-D plane.
3) Based on case (2) and assuming the existence of moving target M with the same coordinate as S_i and not equal to that of R at beam center moment, the relationship of the respective Doppler centroid is $f_{\eta_k} = f_{\mathrm{dc}}(\boldsymbol{\xi}_m) \neq f_{\mathrm{dc}}(\boldsymbol{\xi}_{s_i}) \neq f_{\mathrm{dc}}(\boldsymbol{\xi}_r)$. The offset difference between M and S_i is

$$\Delta\phi_{\mathrm{rcm}}\big(f_{\eta_k=\eta_r}; \boldsymbol{\xi}_{m-s_i}\big) = \frac{\pi}{f_c f_{\mathrm{dr}}(\boldsymbol{\xi}_r)}\big[f_{\mathrm{dc}}^2(\boldsymbol{\xi}_{s_i}) - f_{\mathrm{dc}}^2(\boldsymbol{\xi}_m)\big]. \tag{27}$$

Combined with the aforesaid analysis, the following two conclusions can be easily derived.

1) The stationary targets in the same range cell exhibit linear inclined distribution in the $\eta_k - \kappa$ 2-D domain.
2) The moving target M and the specific stationary target S_i with the same position at the beam center moment have different offsets. It implies that the moving target will be separated from the linearly inclined distribution of the clutter band.

C. Optimal Filter Design for Clutter Suppression

Naturally, the optimal filter for clutter suppression in the η_k–κ domain can be constructed based on the linear inclined

$$\begin{aligned}S_5(\tau, f_{\eta_k}, f_\kappa; \boldsymbol{\xi}_m) &= B\omega_a\left(\frac{f_{\eta_k} - f_{\mathrm{dc}}}{B_a}\right)\omega_\kappa\left(\frac{f_\kappa}{F_\kappa}\right)\exp\big\{j\big[\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m) - \phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_r) f_\kappa - \phi_{\mathrm{src}}(f_{\eta_k}; \boldsymbol{\xi}_r) f_\kappa^2\big]\big\} \\ &\approx B\omega_a\left(\frac{f_{\eta_k} - f_{\mathrm{dc}}}{B_a}\right)\omega_\kappa\left(\frac{f_\kappa}{F_\kappa}\right)\exp\big\{j\big[\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m) - \Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-m}) f_\kappa - \phi_{\mathrm{src}}(f_{\eta_k}; \boldsymbol{\xi}_r) f_\kappa^2\big]\big\}\end{aligned} \tag{22}$$

where

$$\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-m}) = \frac{\pi}{2k_2(\boldsymbol{\xi}_r)}\left[\frac{k_1^2(\boldsymbol{\xi}_r)}{c} - \frac{c}{f_c^2} f_{\eta_k}^2\right]$$

$$S_6(\tau, f_{\eta_k}, \kappa; \boldsymbol{\xi}_m) = B\omega_a\left(\frac{f_{\eta_k} - f_{\mathrm{dc}}}{B_a}\right)\omega_\kappa\left[\kappa - \frac{\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-m})}{2\pi}\right]\exp\left\{j\phi_a(f_{\eta_k}; \boldsymbol{\xi}_m) + j\frac{\pi^2\left[\kappa - \frac{\Delta\phi_{\mathrm{rcm}}(f_{\eta_k}; \boldsymbol{\xi}_{r-m})}{2\pi}\right]^2}{\phi_{\mathrm{src}}(f_{\eta_k}; \boldsymbol{\xi}_r)}\right\} \tag{23}$$

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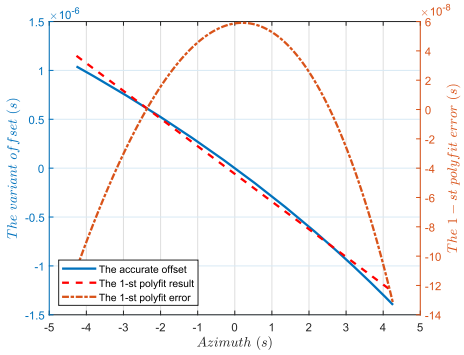


Fig. 6. Offset characteristic of stationary targets in one range cell.

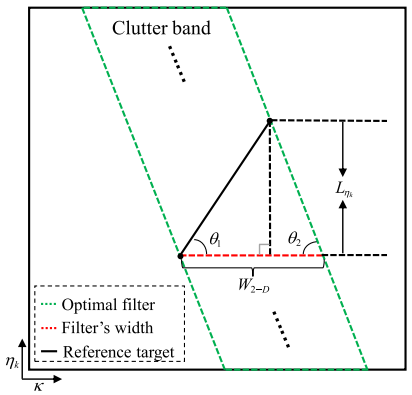


Fig. 7. Schematic for determining the width of the optimal filter.

characteristic of stationary clutter. The steps to construct the optimal filter include the determination of slope and width, and the detailed statement are given as follows.

1) Slope of Optimal Filter: Based on the fact that clutter in the η_k–κ domain is almost linearly inclined, the slope of the corresponding optimal filter needs to be consistent with the characteristic of clutter distribution. The phase difference between stationary targets and reference point targets is $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k};\boldsymbol{\xi}_{r-s_i})$, and the time delay $\Delta\kappa$ is derived from $\Delta\phi_{\mathrm{rcm}}(f_{\eta_k};\boldsymbol{\xi}_{r-s_i})/2\pi$. According to the aforementioned analysis shown in Fig. 6, the time delay $\Delta\kappa$ can be modeled as a first-order polynomial of azimuth time η_k and yields

$$\Delta\kappa = \kappa_0 + \alpha\eta_k \tag{28}$$

where κ_0 is the time delay of reference target and α is the polynomial fitting coefficient.

The slope of the optimal filter on the $\eta_k-\kappa$ plane is determined as α naturally in the light of the aforementioned analysis.

2) Width of Optimal Filter: After obtaining the precise slope for the optimal filter, the width of the corresponding filter is also indispensable. Fig. 7 shows the diagram of determining the corresponding filter width. In Fig. 7, the green dashed line represents the optimal filter to be constructed on the $\eta_k-\kappa$ plane, and the black solid line represents the reference target R. L_{η_k} and W_{2-D} indicate the synthetic aperture length and width of the specific filter, respectively. θ_1 and θ_2 indicate slopes of the reference target and filter, which are given by

$$\begin{cases} \theta_1 &= \frac{\pi}{2} - \arctan\left|k_1(\boldsymbol{\xi}_r)\right| \\ \theta_2 &= \arctan|\alpha|. \end{cases} \tag{29}$$

Consequently, in accordance with the associate geometric relationship, the width of the optimal filter can be represented as

$$W_{2-D} = L_{\eta_k}\left(\frac{1}{\tan\theta_1} + \frac{1}{\tan\theta_2}\right). \tag{30}$$

Combining with the result of slope and width, the optimal filter in the η_k–κ domain can be designed as

$$\Omega_{2-D} = 1 - \mathrm{rect}\left[\frac{\kappa - \alpha\eta_k}{W_{2-D}}\right] \tag{31}$$

where rect represents the rectangular envelope of amplitude 1.

Multiplying $S_7(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m)$ with the optimal filter Ω_{2-D} yields

$$S_8(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m) = S_7(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m)\times\Omega_{2-D}. \tag{32}$$

The signal $S_8(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m)$ after being filtering by the optimal filter has already achieved clutter suppression. Meanwhile, the moving target will be preserved due to the characteristic differences of Doppler parameters.

Next, the 3-D data $S_8(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m)$ after clutter suppression need to be transformed from the τ–η_k–κ domain to the $\tau-\eta_k$ domain. According to the processing of IDR, the corresponding inverse transformation is performed. First, the signal $S_8(\tau,\eta_k,\kappa;\boldsymbol{\xi}_m)$ is operated by 2-D FFT along axes η_k and κ. Then, multiplying $\exp(-j\phi_{\mathrm{rot}}(f_{\eta_k},f_\kappa))$ in the $\tau-f_{\eta_k}$–f_κ domain. Finally, the corresponding 2-D IFFT operation is applied, and the original 2-D signal $S_9(\tau,\eta_k;\boldsymbol{\xi}_m)$ after clutter suppression will be obtained after slicing the middle of the κ-axis.

To further realize the energy accumulation of the moving target and facilitate accurate MTI, FrFT is used to estimate the

$$\begin{aligned}\phi_{\mathrm{rcm}}(f_{\eta_k};\boldsymbol{\xi}_r) &\approx 2\pi\left\{\frac{1}{4k_2(\boldsymbol{\xi}_r)}\left[\frac{k_1^2(\boldsymbol{\xi}_r)}{c} - \frac{c}{f_c^2}f_{\eta_k}^2\right] + \frac{k_3(\boldsymbol{\xi}_r)}{8k_2^3(\boldsymbol{\xi}_r)}\left[\frac{k_1^3(\boldsymbol{\xi}_r)}{c} - \frac{3k_1(\boldsymbol{\xi}_r)c}{f_c^2}f_{\eta_k}^2 - \frac{2c^2}{f_c^3}f_{\eta_k}^3\right]\right.\\ &\quad\left. + \frac{9k_3^2(\boldsymbol{\xi}_r) - 4k_2(\boldsymbol{\xi}_r)k_4(\boldsymbol{\xi}_r)}{64k_2^5(\boldsymbol{\xi}_r)}\left[\frac{k_1^4(\boldsymbol{\xi}_r)}{c} - \frac{6k_1^2(\boldsymbol{\xi}_r)c}{f_c^2}f_{\eta_k}^2 - \frac{8k_1(\boldsymbol{\xi}_r)c^2}{f_c^3}f_{\eta_k}^3 - \frac{3c^3}{f_c^4}f_{\eta_k}^4\right]\right\}\\ &\approx 2\pi\left\{\frac{1}{4k_2(\boldsymbol{\xi}_r)}\left[\frac{k_1^2(\boldsymbol{\xi}_r)}{c} - \frac{c}{f_c^2}f_{\eta_k}^2\right]\right\}\end{aligned} \tag{24}$$

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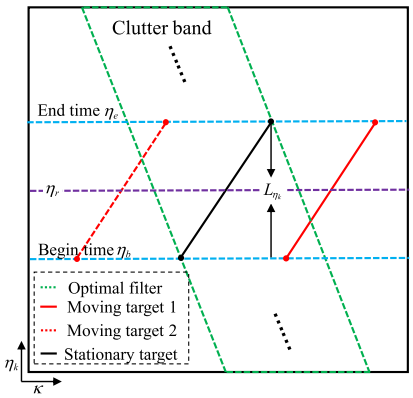


Fig. 8. Schematic for the beam center moment determination of the moving target.

DFR, and the accumulation operation is expressed as follows:

$$S_{10}(\tau,\eta_k;\boldsymbol{\xi}_m)=\underset{f_{\eta_k}}{\mathrm{IFFT}}\left\{\begin{array}{l}\underset{\eta_k}{\mathrm{FFT}}\left[S_9(\tau,\eta_k;\boldsymbol{\xi}_m)\right]\\ \\ \times\exp\left(-j\pi\dfrac{f_{\eta_k}^2}{\hat{f}_{\mathrm{drM}}}\right)\end{array}\right\} \tag{33}$$

where $\hat{f}_{\mathrm{drM}}$ is the estimated DFR. With the above accumulation operation, almost all of the moving target energy can be accumulated. After that, it is easy to realize MTI.

D. Localization and Velocity Estimation of Moving Target

1) Localization: The moving target can be separated from the clutter band in the procedure of IDR. Spontaneously, the beam center moment of moving target η_r can be decided combined with synthetic aperture time and beam irradiation begin/end moment. The schematic is shown in Fig. 8.

In Fig. 8, the red solid line and the red dashed line indicate moving targets in the case of symmetry, the black solid line represents a stationary target, which has the same beam center moment with moving targets, and the interior of 2-D optimal filter marked as green dashed line represents the clutter band. It is clearly found that the start or end location of the moving target is exposed from the clutter band through IDR processing. The higher the velocity, the more the exposed part. Naturally, the beam irradiation begin η_b or end moment η_e (marked as a blue dotted line) can be determined according to this characteristic. Combined with the synthetic aperture time T_s, the beam center moment η_r of the moving target (marked as a purple dotted line) can be determined easily, where $\eta_r = \eta_b + T_s/2$ or $\eta_r = \eta_e - T_s/2$.

On the basis of the beam center moment, the position of the moving target in azimuth dimension can be expressed as $y_m = V_r \times \eta_r$, and then, the coordinate of x_m can be solved by (21). Ultimately, the position coordinates of the moving target (x_m, y_m) can be determined by the closed-form solution without velocity knowledge.

2) Velocity Estimation of Moving Target: In order to estimate 2-D velocities of moving target, parameters of the Doppler centroid frequency f_{dcM} and DFR f_{drM} are applied to establish the related equations (34), as shown at the bottom of the next page. In the processing of energy accumulation of the moving target, FrFT is already used to estimate the DFR f_{drM} of the moving target. In addition, combined with some methods [48], [49] of Doppler ambiguity number estimation, the Doppler centroid frequency can be accurately obtained. Nevertheless, the related equations are not suitable for velocity estimation under conditions that the position is unknown since equations are underdetermined. Fortunately, the position information of the moving target has been carried out in a closed form through the analysis aforementioned. At last, velocities (v_{mx}, v_{my}) can be solved by (34).

We have verified that the closed-form solution of the nonlinear equations can be achieved by software *Wolfram Mathematic*. However, its expression is too complex to describe in this article. Considering the practical applications, Newton's method is suitable for solving nonlinear equations, and the corresponding steps are given as follows:

$$\begin{bmatrix}\frac{\partial f_1}{\partial v_{mx}}\Big|_{v_{mx}=v_{mx}^k}, & \frac{\partial f_1}{\partial v_{my}}\Big|_{v_{my}=v_{my}^k}\\ \\ \frac{\partial f_2}{\partial v_{mx}}\Big|_{v_{mx}=v_{mx}^k}, & \frac{\partial f_2}{\partial v_{my}}\Big|_{v_{my}=v_{my}^k}\end{bmatrix}\begin{bmatrix}v_{mx}^{k+1}\\ \\ v_{my}^{k+1}\end{bmatrix}=\begin{bmatrix}f_{\mathrm{dcM}}\\ \\ f_{\mathrm{drM}}\end{bmatrix}. \tag{35}$$

Therefore, the iteration procedure can be also expressed as

$$\begin{bmatrix}v_{mx}^{k+1}\\ \\ v_{my}^{k+1}\end{bmatrix}=\begin{bmatrix}\frac{\partial f_1}{\partial v_{mx}}\Big|_{v_{mx}=v_{mx}^k}, & \frac{\partial f_1}{\partial v_{my}}\Big|_{v_{my}=v_{my}^k}\\ \\ \frac{\partial f_2}{\partial v_{mx}}\Big|_{v_{mx}=v_{mx}^k}, & \frac{\partial f_2}{\partial v_{my}}\Big|_{v_{my}=v_{my}^k}\end{bmatrix}^{-1}\begin{bmatrix}f_{\mathrm{dcM}}\\ \\ f_{\mathrm{drM}}\end{bmatrix} \tag{36}$$

where (v_{mx}^k, v_{my}^k) is the current position of iteration point and $(v_{mx}^{k+1}, v_{my}^{k+1})$ is the next position of iteration point. $[\bullet]^{-1}$ represents the generalized inverse of matrix.

Through the establishment of equations with respect to f_{dcM}, f_{drM}, Newton's method is applied to achieve the iterative solution. Eventually, 2-D velocities of the moving target can be precisely solved.

E. Performance Analysis

1) Value of Q: The sample number Q of the κ-axis is related to the size of a 3-D data cube after increased dimension. To minimize computational burden, it is indispensable to find the optimal range of Q. Essentially, Q is determined by synthetic aperture time

$$Q = T_s \times F_\kappa \tag{37}$$

where F_κ is the sampling rate of the κ-axis. In the following simulation, F_κ is set to be consistent with the PRF.

2) Minimum Detectable Velocity: Through the above analysis, it can be found that the separation degree between the moving target and the clutter band is determined by linear phase difference on the κ-axis. The greater the velocity, the greater the degree of separation. When the velocity is too small, a significant portion of energy of the moving target will be submerged in the clutter band. According to the proposed method, the moving target's energy in the clutter band is also suppressed, so the energy of the moving target will suffer loss to a certain degree. Next, the critical detectable schematic of the moving target is shown in Fig. 9, where the critical

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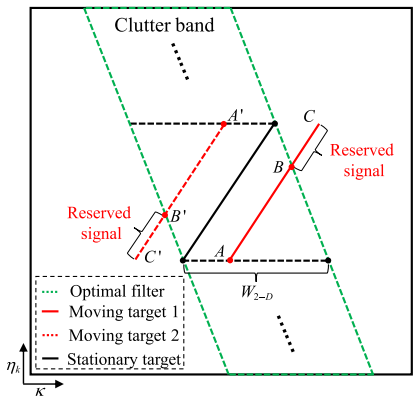
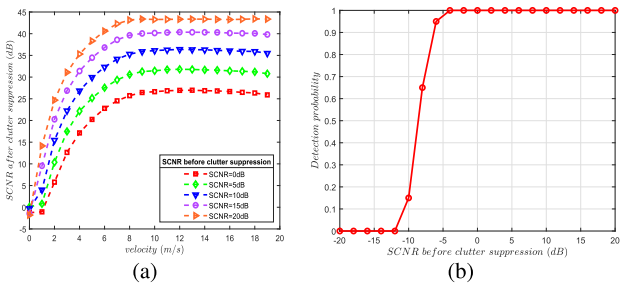


Fig. 9. Critical detectable schematic of the moving target.



(a) (b)

Fig. 10. Variation curve of SCNR with velocity and the detect ability with different SCNRs. (a) Variation curve of detection SCNR with velocity. (b) Detection ability with different SCNRs.

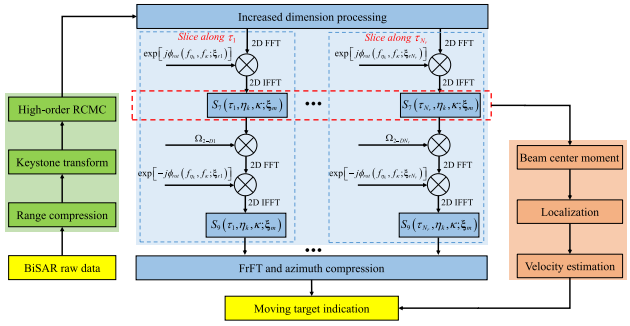


Fig. 11. Block diagram of the proposed method.

detectable situation means that SCNR after clutter suppression is equal to 13 dB [13], [44].

Fig. 9 represents the signal on the η_k–κ plane after IDR processing. The green dashed line is the designed optimal filter for clutter suppression, and the width marked as the black dashed line is notated as W_{2-D}. The black solid line represents a stationary target, and the red solid line AC and the red dashed line $A'C'$ represent moving targets in symmetry. The significant portion of energy of moving targets (AB and $A'B'$) is submerged in the clutter band because of the small velocity. After clutter suppression by the proposed method, only the signal exposed outside the clutter band (BC and $B'C'$) can be reserved. From [13] and [45], the SCNR should be higher than 13 dB in order to realize reliable detection. Supposing that the critical situation shown in Fig. 9 exactly satisfies the detection threshold of 13 dB, then we want to explore the detectable velocity in this critical case. The following Monte Carlo simulation draws the variation curve of detection SCNR with velocity after clutter suppression under different SCNRs before clutter suppression.

In Fig. 10(a), Monte Carlo simulation is carried out, where 100 times Monte Carlo experiments are performed for each variation. As a consequence, the detection SCNR after clutter suppression will be greater than 13 dB when the velocity is higher than 4 m/s. The higher SCNR before clutter suppression, the greater detection SCNR after clutter suppression. Therefore, it is not difficult to conclude that, when SCNR before clutter suppression is greater than or equal to 0 dB, the minimum detectable velocity is 4 m/s.

Furthermore, to evaluate the performance of the proposed method with different SCNRs, the corresponding Monte Carlo simulation is also performed, where the false alarm probability is set as 10^{-4}. The scope of tested SCNR is set from -20 to 20 dB. Besides, the velocity of the moving target is set as 8 m/s because the SCNR after clutter suppression tends to be stable from Fig. 10(a). It can be easily drawn the conclusion that the proposed method can obtain a reliable detection performance when SCNR ≥ -4 dB.

3) Computational Complexity: To better present the proposed method, the block diagram of the proposed algorithm is given first.

The proposed method includes RCMC, IDR-related processing, and parameters' estimation. The mainly computational cost mainly focuses on RCMC and IRD-related processing. RCMC processing mainly contains range compression ($2N_aN_r\log(N_r) + N_aN_r$), KT ($(2(2M_k-1)N_aN_r$), and high-order RCMC ($2N_aN_r\log(N_r) + N_aN_r$), where N_a and N_r denote the number of azimuth and range dimensions and M_k represents the number of neighbor samples used for the azimuth and range interpolation. For the one slice in increased dimension-related processing, its major implementation procedures focus on 2-D FFT ($2N_aQ\log(Q) + 2N_aQ\log(N_a)$), multiplying rotation phase factor (N_aQ), 2-D

$$\begin{cases} f_1: f_{\mathrm{dcM}} = -\dfrac{v_{mx}(x-X_R)+\left(V_r-v_{my}\right)(Y_R-y)}{\lambda\sqrt{(X_R-x)^2+(Y_R-y)^2+Z_R^2}} - \dfrac{(V_{tx}-v_{mx})(X_T-x)+\left(V_{ty}-v_{my}\right)(Y_T-y)}{\lambda\sqrt{(X_T-x)^2+(Y_T-y)^2+Z_T^2}} \\ f_2: f_{\mathrm{drM}} = -\dfrac{\left[V_{r,my}(x-X_R)+v_{mx}(y-Y_R)\right]^2+\left(v_{mx}^2+V_{r,my}^2\right)Z_R^2}{\lambda\left[(X_R-x)^2+(Y_R-y)^2+Z_R^2\right]^{3/2}} - \dfrac{\Gamma^2+\left(V_{t,mx}^2+V_{t,my}^2\right)Z_T^2}{\lambda\left[(X_T-x)^2+(Y_T-y)^2+Z_T^2\right]^{3/2}} \end{cases}$$

where

$$\Gamma = V_{t,my}(x-X_T)+V_{t,mx}(Y_T-y) \tag{34}$$

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TABLE I
SIMULATION PARAMETERS

Parameters	Symbol	Value
Carrier Frequency	f_c	9.6GHz
LFM Bandwidth	B_r	120MHz
Pulse Repetition Frequency	PRF	1200Hz
Synthetic Aperture Time	T_s	1s
Transmitter Position	T	(-5000 -1000 3000)m
Receiver Position	R	(-500 -6000 4000)m
Transmitter Velocity	V_t	(-50 150 0)m/s
Receiver Velocity	V_r	(0 200 0)m/s

IFFT $(2N_aQ\log(Q) + 2N_aQ\log(N_a))$, filtering by optimal filter (N_aQ), multiplying inverse rotation phase factor (N_aQ), FrFT, and azimuth compression $(PN_a\log(N_a) + 2N_a\log(N_a))$, where Q and P indicate the sample number of κ dimension and the iteration number of FrFT. Therefore, the total numbers of real-floating operations are

$$4N_aN_r\log(N_r) + 4M_kN_aN_r + N_r\begin{bmatrix}4N_aQ\log(Q) + 4N_aQ\log(N_a) + \\ 3N_aQ + (P+2)N_a\log(N_a)\end{bmatrix}. \quad (38)$$

It is worth mentioning that the related processing of all slices in the IDR procedure is independent of each other so that the processing can be accelerated in parallel. In other words, there is no more time spent on cyclic processing with enough memory space. Consequently, the computational complexity of per slice processing is of order $O(N^2\log(N))$, which is equal to that of the two-step [7] method and the Doppler filtering method [10], where N is the 1-D size of the data.

IV. NUMERICAL SIMULATIONS

To prove the effectiveness of the joint clutter suppression and MTI method, two simulation cases are demonstrated here, in which one is a point target simulation case and the other is an area scene simulation case. The associated system parameters are shown in Table I. Meanwhile, it is worth mentioning that the relevant simulation parameters in this article are all in accordance with Table I.

A. Case I

This section is intended to explain the processing flow of the proposed method in detail. As shown in Fig. 12, there are stationary targets $S1$–$S7$ in the scenario. Their positions are $(-750.80, 583.33)$ m, $(-509.48, 416.67)$ m, $(-291.59, 250.00)$, (0,0) m and $(556.72, -583.33)$ m, $(412.60, -416.67)$ m, $(257.19, -250.00)$ m, respectively. This scenario also exists moving target $M1$ with velocity (5, 6) m/s simultaneously, and the corresponding position at azimuth zero moment is $(-397.90, 333.33)$ m.

Fig. 12(a) shows the distribution characteristics of moving target $M1$ and stationary targets $S1$–$S7$. The echo data before and after RCMC with the first-order KT and high-order RCMC are shown in Fig. 12(b) and (c). These targets fall in the same range cell after RCMC, and moving target $M1$ marked

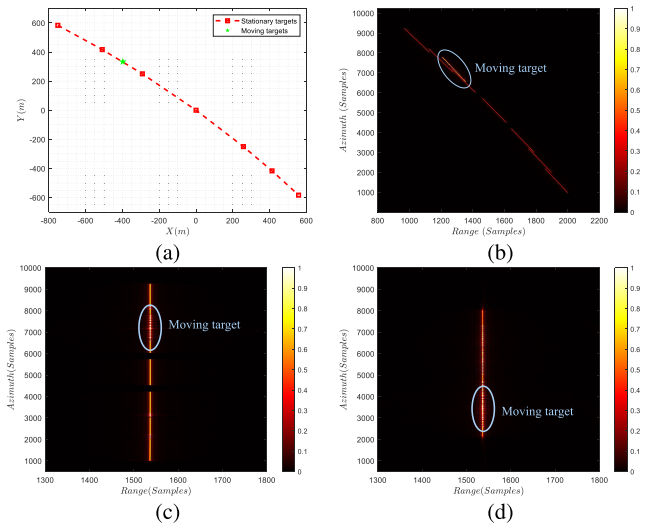


Fig. 12. Position distribution and echo characteristics of stationary targets and moving targets. (a) Position distribution of stationary targets and moving targets. (b) Signals before RCMC. (c) Signals after RCMC. (d) Doppler spectrum.

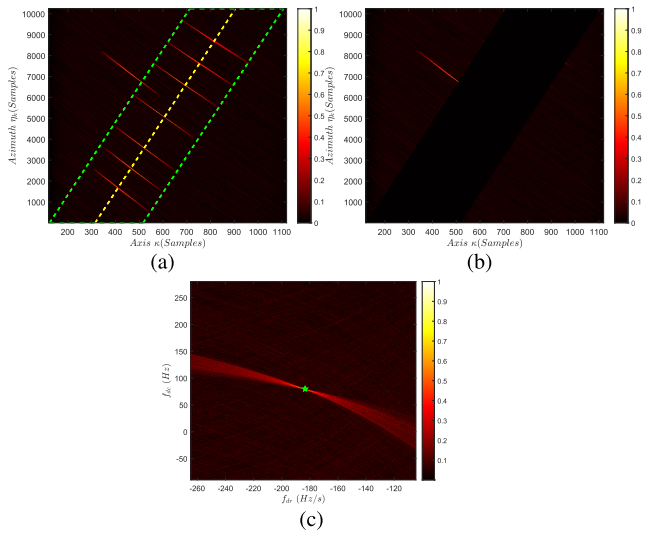


Fig. 13. Separation procedure of moving target from stationary clutter with IDR processing. (a) Signal after IDR processing in the η_k–κ domain and 2-D optimal filter for clutter suppression. (b) Residual moving target signal after clutter suppression. (c) FrFT detection result after inverse IDR processing.

with the wathet blue circle overlaps with stationary targets. After the subsequent azimuth FFT, the RD spectrum is shown in Fig. 12(d). It is obvious that moving target $M1$ is fully submerged into the clutter spectrum. Therefore, the traditional Doppler filtering method [10] in monostatic SAR is not suitable for MTI in BiSAR.

With IDR processing as (20)–(25), these targets in the same range cell are rotated to the η_k–κ domain. From Fig. 13(a), it can be clearly observed that stationary target signals in the η_k–κ domain present a linear inclined distribution. It confirms the analysis aforementioned in Fig. 6. On the other hand, because of the Doppler centroid difference caused by the moving target's velocity, the moving target is exposed outside the stationary clutter band, which is in accordance with the prospective analysis for (27). Spontaneously, the optimal filter marked as the green dashed line in Fig. 13(a) on the η_k–κ plane

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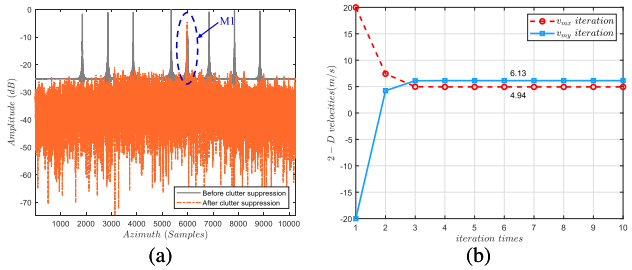


Fig. 14. Moving target detection result before and after clutter suppression, and Newton's iteration procedure of moving target's velocity. (a) Comparison result before and after clutter suppression. (b) Newton's iteration procedure of moving target's velocity.

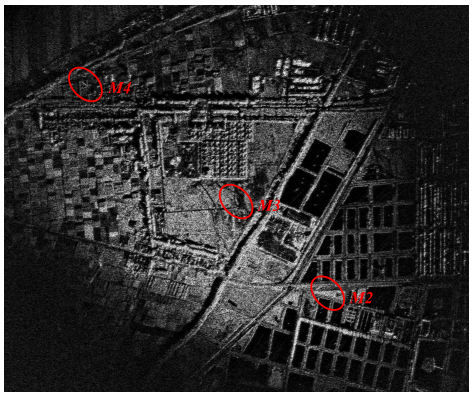


Fig. 15. Imaging result with clutter background and moving targets.

is constructed to distinguish stationary clutter and moving target. The corresponding slope and width of the optimal filter can be calculated based on (28) and (30), respectively. The yellow dashed line in Fig. 13(a) represents the first-order polynomial fitting of stationary targets' offset difference compared with the reference stationary target on the κ-axis, and the slope of the optimal filter is, thereby, determined. Fig. 13(b) indicates the result after clutter suppression, where only little stationary clutter and noise are reserved. With corresponding inverse IDR processing and FrFT operation, the DFR of the specific moving target could be estimated (marked as a green pentagram), with the result after FrFT shown in Fig. 13(c).

To detect the moving target precisely, the moving target is accumulated in the azimuth time domain on the basis of (33) and DFR estimation result. The accumulation result is shown in Fig. 14(a). The gray solid line represents the accumulation result before clutter suppression, and the specific moving target $M1$ is fully submerged because of strong stationary clutter. The orange dashed line indicates the accumulation result after clutter suppression. The stationary clutter is suppressed greatly, and the SCNR has improved by 20 dB approximately so that an MTI of $M1$ can be realized easily and precisely.

Furthermore, to realize moving target positioning and velocity estimation, the specific implementation should start with the determination of the beam center moment in accordance with the processing flow. From Fig. 13(a) and (b), the beam end moment can be easily acquired. According to the synthetic aperture time $T_s = 1$ s, the beam center moment of the specific moving target is determined as 1.68 s. Then, the position coordinates of the moving target are solved as

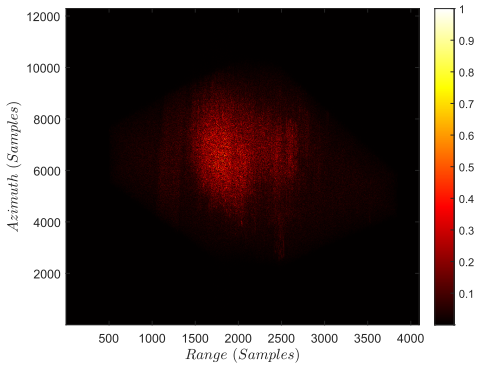


Fig. 16. RD spectrum after RCMC of scene targets and moving targets.

$(-400.07, 335.00)$ m according to (21), and the position coordinates error is 0.53%. Next, velocity estimation results can be solved by Newton's method in (34), whose iteration process is shown in Fig. 14(b). To solve (34), the Doppler centroid frequency and DFR should be determined first. Fortunately, the DFR has been estimated through FrFT. The Doppler centroid can be also easily gained combined with some Doppler ambiguity number estimation methods [48], [49]. The initial value for the Newton iteration is set to be $(20, -20)$ m/s here, and the converged velocity estimation results can be acquired in less than five iterations. This confirms the practical feasibility of this method. The converged velocity results are $(4.94, 6.13)$ m/s, and the velocity estimation error is 1.83%. Through the simulation above, the positioning error and the velocity estimation error are both minor, a conclusion can be drawn that accurate positioning, and velocity estimation of the moving target can be achieved.

B. Case II

To further substantiate the effect of the proposed method, the simulation case about the area scenario is demonstrated in the following. The system parameters remain in accordance with Table I. There are three moving targets at the same range cell, noted as $M2$–$M4$. The positions of moving targets $M2$–$M4$ at azimuth zero moment are $(-750.80, 583.33)$, $(-291.58, 250)$, and $(412.60, -416.67)$ m, and corresponding velocities are $(-2, 8)$, $(4, 6)$, and $(-3, -4)$ m/s, respectively. The imaging result with scene clutter is shown in Fig. 15. The imaging size is about 2×2 km. The imaging area presents strong nonhomogeneous because of the complex terrain (e.g., buildings, farmland, and ponds). Three moving targets mentioned above are identified in the imaging result by red ellipses. Nevertheless, due to the influence of strong clutter, moving targets are completely masked by a clutter background. Therefore, it is greatly difficult to realize MTI in the imaging domain directly.

The corresponding RD spectrum after RCMC is shown in Fig. 16. It is clear that the clutter spectrum is extended severely so that the RD spectrum of moving targets is completely covered by the spectrum of strong clutter. Naturally, the Doppler filtering-based methods will fail to indicate moving targets.

According to the method proposed in this article, the distribution of signals in the η_k–κ domain will be presented in

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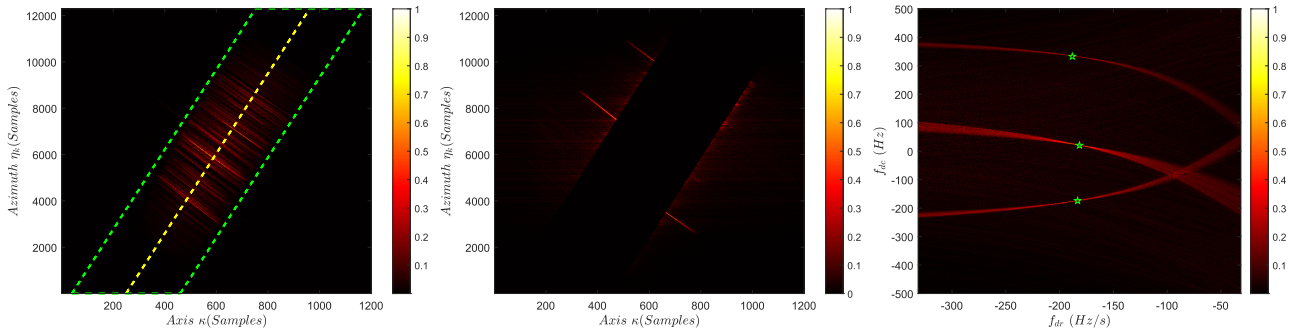


Fig. 17. Procedure of clutter suppression and MTI. (a) IDR processing with one special range cell. (b) Residual signal after clutter suppression. (c) FrFT result after inverse rotation operation.

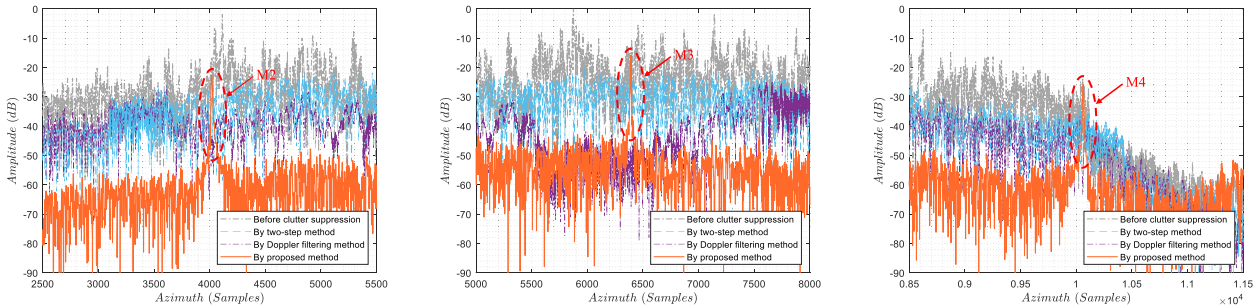


Fig. 18. Azimuth profile accumulation of different moving targets before and after clutter suppression. (a) Azimuth profile of $M2$. (b) Azimuth profile of $M3$. (c) Azimuth profile of $M4$.

Fig. 17 after IDR processing. In Fig. 17(a), the yellow dashed line represents the first-order polynomial fitting of time delay on the κ-axis between stationary clutter and reference target, and the green dashed line represents the designed optimal filter in the η_k–κ domain. Thereafter, the residual signal is shown in Fig. 17(b), where almost all clutter is effectively suppressed. According to the processing flow of the proposed method, the corresponding inverse IDR processing and the FrFT operation are applied. The FrFT result is shown in Fig. 17(c); it could easily get the Doppler parameters of respective moving targets (marked as a green pentagram), laying the foundation for subsequent accumulation and position, as well as velocity parameters' estimation.

After clutter suppression, the strong clutter is greatly suppressed. The indication result of moving targets is shown in Fig. 18 with subsequent accumulation. Before clutter suppression, moving targets are completely submerged in the clutter signal. The SCNR has been improved to about 31.30, 35.67, and 27.77 dB via clutter suppression by the proposed method, respectively. Because the detected SCNR is much higher than 13 dB, after clutter suppression, MTI in BiSAR can be effectively conducted by the proposed method. Besides, Fig. 18 presents the comparison result between the proposed method and the two-step method [7], as well as the Doppler filtering method [10]. The two comparison methods cannot improve the SCNR effectively, so accurate MTI cannot be realized. Essentially, the two-step method utilizes two subapertures with the same Doppler history for stationary targets and different Doppler histories for moving targets to achieve MTI, which is not applicable to BiSAR because of distinct Doppler characteristics. For the Doppler filtering method, it cannot achieve MTI when the Doppler spectrum of the moving target is submerged in that of stationary clutter.

To display the performance of the proposed method more perspicuously, the comparison imaging results are illustrated in Fig. 19. From Fig. 19(a), it can be found that only a few clutter is suppressed by the two-step method, where three moving targets are cloaked in the remaining strong clutter background. Thus, accurate MTI is still unrealistic at this stage. The imaging result after Doppler filtering is indicated in Fig. 19(b). Similarly, the clutter is not completely suppressed, and the moving target in the passband of corresponding Doppler filtering is also suppressed. As a result, moving targets are still difficult to indicate. Comparatively, the imaging result of the proposed method is demonstrated in Fig. 19(c). It is obvious that the clutter is fully suppressed so that moving targets can be indicated easily and accurately.

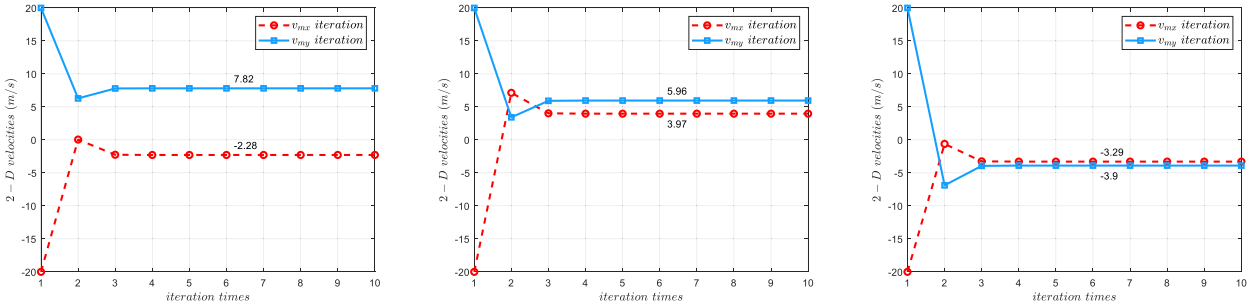
Next, to realize precise positioning of moving targets under the condition of unknown velocity parameters, the respective beam center moment is determined first associated with the signal begin/end moment and synthetic aperture time. From Fig. 17(a) and (b), the begin/end moment is easy to obtain extremely. The values of moving targets' beam center are -2.07, 1.26, and 2.92 s, and then, the position coordinates are (412.60, -416.67), (-291.59, 250.00), and (-750.81, 583.33) m by solving (21). Thereafter, the position errors are 0.5%, 0.48%, and 0.28%. It implies that the proposed method can achieve high-precision positioning of moving targets.

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(a) (b) (c)

Fig. 19. Imaging result after different clutter suppression methods. (a) Two-step method in [7]. (b) Doppler filtering method in [10]. (c) Proposed method.



(a) (b) (c)

Fig. 20. 2-D velocities iterative solving process of moving targets by Newton's method. (a) Velocity iterative solution curve of $M2$. (b) Velocity iterative solution curve of $M3$. (c) Velocity iterative solution curve of $M4$.

What is more, velocity equations shown in (34) are established to solve the 2-D velocities of moving targets. The Doppler parameters can be obtained by FrFT and Doppler ambiguity number estimation methods; there will be no more specific discussion here. Newton's method iterative solution process is shown in Fig. 20. After less than five iterations, velocities of moving targets $M2$–$M4$ converge to $(-2.28, 7.82)$, $(3.97, 5.96)$, and $(-3.29, -3.90)$ m/s, respectively. Compared with the theoretical velocity, velocity solution errors are 4.04%, 0.69%, and 6.14%. Owing to the precise prior knowledge about bistatic platforms' positions and velocities, velocity solution error of moving targets is mainly caused by the estimation accuracy of Doppler parameters. The more accurate the Doppler parameters' estimation, the more precise the velocity solution results.

In summary, the proposed method is a prerequisite to enable MTI by filtering in the newly constructed 2-D domain. Fortunately, the construction of the optimal filter does not need to consider the effect of nonhomogeneous clutter. Besides, the proposed method designs the corresponding optimal filter for each range cell, ignoring the influence of nonhomogeneous and nonstationary characteristics for different range cells. Therefore, the proposed method has good robustness.

V. Conclusion

In this article, based on single-channel BiSAR, a joint clutter suppression and MTI method is proposed. The main innovation of the proposed method is to separate moving targets from the stationary clutter in a newly constructed 2-D ARTD. First, RCM and Doppler spectrum extension characteristics are analyzed in detail. Then, the joint clutter suppression and MTI method is described, which mainly involves four steps, i.e., KT operation, IDR processing, clutter suppression via optimal filter, and parameters' estimation. One of the most critical steps is IDR processing, which constructs 2-D ARTD and separates moving targets from stationary clutter by specific rotation. The main advantages of the proposed method include the following: 1) the single-channel system has lower complexity compared with the multichannel system, and it is convenient for practical engineering applications; 2) the clutter suppression performance is not affected by the geometrical configuration-dependent clutter characteristic, and thus, accurate MTI can be achieved; and 3) the positioning and velocity estimation of moving targets can be realized simultaneously without additional heavy computation. From the simulation results, it is proven that the proposed method can achieve reliable indication when SCNR ≥ -4 dB. Meanwhile, numerical simulations also confirm the robustness and effectiveness of the proposed method.

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